KATLYN L. HOBBS

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Stellar Activity; Exoplanet Detection Methods; Radial Velocity
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EDUCATION

PhD, Astrophysics - Queen's University Belfast, UK *Sep 2023 - present*
Thesis title: *Modelling the Impact of Stellar Activity on the Detection of Exoplanets*
Supervisors: Prof. Christopher A. Watson & Dr. Ernst J. W. de Mooij
Expected completion: June 2027

BASc, Engineering Physics - Queen's University, Canada *Oct 2017 - Jul 2022*
Specialization in Materials Engineering
Undergraduate Thesis: *Investigating Magnetic Field Diagnosis in Preparation for the Polstar High-Resolution Spectropolarimetry MIDEX Mission*
Supervisor: Prof. Gregg A. Wade

RESEARCH POSITIONS

Pre-Doctoral Fellowship - Flatiron Institute (CCA), New York, USA *Sep 2025 - Jan 2026*
Project: *Characterizing Novel Stellar Activity Indicators using NEID, KPF, HARPS-N, and EXPRES Sun-as-star observations*
Supervisors: Dr. Megan Bedell & Dr. Ryan Rubenzahl

RESEARCH HIGHLIGHTS

- **Applying Spectral Ratio Analysis (SRA)** to HARPS-N, KPF, NEID, and EXPRES Sun-as-a-star observations to isolate spectral variations induced by active regions such as faculae, leveraging Solar Dynamics Observatory (SDO) data for comparison with disk-resolved measurements
- Comparing **PHOENIX and MuRAM synthetic models to observed SRA residuals** to estimate faculae filling factor and provides constraints on facular properties directly from spectra
- Applying **Principal Component Analysis (PCA) to SRA residuals** across hundreds of lines to identify the dominant line-shape changes driven by stellar activity

SELECTED TALKS

GPRV, Oxford, UK *Sep 2026*
Contributed talk: *Tracing the Spectral Fingerprints of Magnetically Active Regions Across the Optical Spectrum*

HORS3S, Granada, Spain *Jul 2026*
Contributed talk: *Tracing the Spectral Fingerprints Sunspots and Plage Across The Optical Spectrum*

Exoplanets 6, Porto, Portugal *Jul 2026*
Contributed talk: *Tracing the Spectral Fingerprints of Magnetic Activity Using Sun-as-a-star Observations*

Cool Stars 23, Tokyo, Japan *Jun 2026*
Contributed talk: *Tracing the Spectral Fingerprints of Magnetic Activity Using Sun-as-a-star Observations*

UK Exoplanet Meeting 2025, Leeds, UK *Apr 2025*
Contributed talk: *Modeling the Impact of Stellar Activity on HARPS-N Solar Spectra*

SUMMER SCHOOLS AND WORKSHOPS

La Silla Observing School (ESO – Santiago & La Silla, Chile) *Feb 2025*
Two-week Summer School on Observational Astronomy

- Attended lectures and practical sessions on observing strategies, instrumentation, proposal writing, and data reduction.
- Carried out an observing project on HD 86081 b, including data collection over three nights at La Silla, data reduction using the ESO pipeline, and analysis to measure radial velocities and derive orbital period and companion mass.

CodeAstro (CIERA, Northwestern University – Evanston, IL) *Jul 2024*
Workshop – Coding for Astronomy

- Participated in a week-long intensive program focused on best practices in scientific coding, version control with Git, and parallelization/multiprocessing.
- Completed a Git-based group project and developed a pip-installable Python package for downloading and reducing HARPS-N and HARPS spectra.

TEACHING EXPERIENCE

Sep 2026 - present – Marker and Teaching Assistant for Maths for Engineers.

Sep 2024 - Feb 2025 – Teaching assistant for Physics laboratories.

Sep 2023 - Apr 2024 – Teaching assistant for undergraduate Python programming course.

Sep 2021 - Apr 2022 – Marker and teaching assistant for Maths and Physics for Engineers course.

OUTREACH AND SERVICE

ARC Equitea Treasurer *Nov 2023 - present*
Treasurer of ARC Equitea (initiative providing a forum to discuss ED&I issues in academia and develop possible solutions). Co-wrote grant application and was awarded £300 by Queen's University Belfast Student Union. Gave presentations on topics including impostor syndrome, unconscious bias, and the challenges and benefits of moving abroad for academia.

QUB Astronomy Day 2023/2024 – NI Science Festival *Feb 2023/2024*
Created materials and ran activities for outreach day.

SCHOLARSHIPS AND FUNDING AWARDED

RAS Grant *Jun 2026*
£1000 GBP to support travel to attend Exoplanets 6 Conference in Porto, Portugal

ESO Financial Support *Feb 2025*
$600 USD to support travel to attend La Silla Observing School in Santiago, Chile

Emily Sarah Montgomery Travel Scholarship *Jul 2024*
£400 GBP to support travel to attend Code/Astro workshop 2024 in Chicago, IL, USA

PUBLICATIONS

First author:

1. **Katlyn L. Hobbs**, Christopher A. Watson, Jean C. Costes et al. 2026, *The Fe I 4377 Å line as a solar faculae indicator: insights from spectral ratio analysis*, 2026, MNRAS, Vol. 551, Issue 2
   doi:10.1093/mnras/stag1462
   Contribution: Led the analysis, interpretation of results, and writing of the manuscript.

2. **Katlyn L. Hobbs**, Ryan Rubenzahl, Megan Bedell et al. *Cross-Instrument Solar Activity Signatures: PCA of Sun-as-a-Star Line Profiles from HARPS-N, KPF, NEID, and EXPRES*, in preparation
3. **Katlyn L. Hobbs**, Christopher A. Watson et al. *Tracking the Radial Velocities of Solar Faculae with Spectral Ratio Analysis*, in preparation

Contributing author:

1. Jean C. Costes, Christopher A. Watson, Ernst de Mooij, **Katlyn L. Hobbs** et al. 2026, *Spectral Ratio Analysis: probing of a new suite of stellar activity indicators as a tool for astrophysical noise mitigation*, MNRAS, Vol. 550, Issue 2
   doi:10.1093/mnras/stag1173
   Contribution: Co-developed the Spectral Ratio Analysis pipeline.

2. Niamh Mallaghan, Ernst de Mooij, Christopher Watson et al. (including **Katlyn L. Hobbs**, 2026, *NGTS clusters survey - VII. An Enigmatic Short-Period Circumsecondary Disk Candidate in Orion*, MNRAS, Vol. 552, Issue 3
   doi:10.1093/mnras/stag1802
   Contribution: Assisted with analysis of spectroscopic data and provided feedback on manuscript.

3. Toby Rodel, Solène Ulmer-Moll, Samuel Gill, et al. (including **Katlyn L. Hobbs**) 2025, *TIC-65910228b/NGTS-38b, a 180d transiting warm super-Jupiter*, MNRAS, Vol. 549, Issue 4
   doi:10.1093/mnras/stag1061
   Contribution: Provided feedback on manuscript.

4. Toby Rodel, Christopher A. Watson, Solène Ulmer-Moll et al. (including **Katlyn L. Hobbs**) 2025, *NGTS-EB-7, an eccentric, long-period, low-mass eclipsing binary*, MNRAS, Vol. 537, Issue 1, Pages 35-55
   doi:10.1093/mnras/stae2799
   Contribution: Provided feedback on manuscript.

5. Sean M. O'Brien, Megan E. Schwamb, Christopher A. Watson et al. (including **Katlyn L. Hobbs**) 2025, *NGTS-EB-8: A double-lined eclipsing M+M binary discovered by citizen scientists*, AJ, 170, 316
   doi:10.3847/1538-3881/ae0e0d
   Contribution: Assisted with analysis of spectroscopic data and provided feedback on manuscript.

6. Folsom, C.P., Ignace, R., Erba, C. et al. (including **Katlyn L. Hobbs**) 2022, *Ultraviolet spectropolarimetry: investigating stellar magnetic field diagnostics.*, Astrophys Space Sci 367, 125
   doi:10.1007/s10509-022-04140-8
   Contribution: Generated synthetic Stokes I and V profiles from stellar atmosphere models across a range of stellar temperatures, metallicities, rotation rates, and magnetic field strengths. Identified the optimal spectral resolution and signal-to-noise requirements necessary to detect Zeeman signatures in hot star spectra, informing the capabilities, requirements, and feasibility of Polstar. The results of this analysis form the basis for Section 2 of the publication.